European Option Pricing under the Lin–Lin–He Stochastic Volatility Model

Rami Younes

Supervised by Prof. Fabian Bastin
Université de Montréal

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1 Introduction

This report presents a numerical study of European option pricing under the improved Stein–Stein stochastic volatility model proposed by Lin, Lin & He, hereafter referred to as the LLH model. The model extends the Schöbel–Zhu (1999) framework by introducing a time-varying long-run mean θ_t driven by a geometric Brownian motion, adding two parameters λ (drift) and η (vol-of-vol) to the volatility dynamics. Schöbel–Zhu model improves upon the Stein-Stein model by allowing correlated Brownian motions that govern returns and volatility.

A semi-analytical European pricing formula is derived in Lin , Lin &He (2024) via characteristic functions, computed by solving a system of six coupled ODEs numerically. Following the example of Lin, Lin, & He (2024), we test the validity of our implementation of the formula by:

1. comparing it against Monte Carlo estimates across moneyness and maturity scenarios;
2. testing edge-case recovery of Black–Scholes, Stein–Stein, and Schöbel–Zhu prices;
3. studying the sensitivity of the price to the LLH extension parameters λ and η.

All experiments are repeated under two reference parameter sets (Tables 1 and 2 from Lin, Lin & He) to assess robustness across calibration regimes.

Implementation Note.
The ODE system for the characteristic function coefficients C, D, E, F, G, H must be *autonomous*: the right-hand side depends only on the current state, not on the time to maturity τ. This corrects Lin, Lin & He paper's presentation, which suggests τ-dependence through $\partial D/\partial\tau$ notation. The implementation under the autonomous condition is validated by close agreement with independent Monte Carlo estimates and consistency through out the performed experiments.

2 The LLH Model

The LLH model specifies the following system of stochastic differential equations under the risk-neutral measure:

$$\frac{dS_t}{S_t} = r\,dt + \sigma_t\,dW_t^1, \tag{1}$$

$$d\sigma_t = \kappa(\theta_t - \sigma_t)\,dt + \nu\,dW_t^2, \tag{2}$$

$$d\theta_t = \lambda\,dt + \eta\,dB_t, \tag{3}$$

$$dB_t = B_t\,dW_t, \quad B_0 = 1, \tag{4}$$

where $\mathrm{Cov}(W_t^1, W_t^2) = \rho\,t$ and W_t is an independent standard Brownian motion. The unobserved parameter set is $\Theta = (\kappa, \nu, \lambda, \eta, \rho, \sigma_0, \theta_0)$.

Two reference parameter sets from Lin, Lin & He (2024) are used throughout:

	r	κ	ν	λ	η	ρ	σ_0	θ_0
LLH Table 1	0.01	5.0	0.2	0.9	0.01	−0.2	0.15	0.18
LLH Table 2	0.01	4.9394	0.3943	0.3115	0.4112	0.1691	0.2924	0.1319

Table 1: Reference parameter sets.

3 Semi-Analytical Pricing Formula

The European call price at time t with strike K and maturity T is

$$U(\tau;\, S_t, K) = S_t\, P_1(\tau; K) - K\, e^{-r\tau}\, P_2(\tau; K), \tag{5}$$

where $\tau = T - t$ and the probabilities P_j are recovered from the characteristic functions f_j via Fourier inversion:

$$P_j = \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \mathrm{Re}\left[\frac{e^{-i\varphi \ln K}}{i\varphi}\, f_j(\tau; \varphi)\right] d\varphi, \quad j = 1, 2. \tag{6}$$

The characteristic function takes the exponential-quadratic form

$$f_j(\tau; \phi) = \exp\Big(C + D\,\sigma_t^2 + E\,\sigma_t\theta_t + F\,\theta_t^2 + G\,\theta_t + H\,\sigma_t + i\phi\, x_t\Big), \tag{7}$$

where $x = \ln S_t$ and the coefficients (C, D, E, F, G, H) satisfy a coupled system of six autonomous ODEs with initial conditions $C(0, \phi) = D(0, \phi) = \cdots = H(0, \phi) = 0$, integrated from $\tau = 0$ to the desired maturity using a fourth-order Runge–Kutta scheme. The integral (6) is evaluated on a uniform φ-grid via the trapezoid rule.

European put prices are obtained via put-call parity:

$$P = C - S + K\, e^{-r\tau}. \tag{8}$$

4 LLH Formula vs Monte Carlo

We compare the LLH formula and Monte Carlo prices across moneyness for spot values S_0 $90, 95, 100, 105, 110$ and strikes representing ITM ($K = 80$), ATM ($K = 100$), and OTM ($K = 120$) regimes. The time to maturity is 1 year and the number of MC paths is 10^5 with 52 steps.

We observe that for LLH Table 1 parameter set, the formula and MC prices agree closely to within 2% for in ITM and ATM options. Relative error is within 3% for the OTM options, where the option value is smaller and discretisation error becomes more significant. For the LLH Table 2 parameters, the relative error is within 9% for OTM options as opposed to the ITM options that remain more consistent to within 3%.

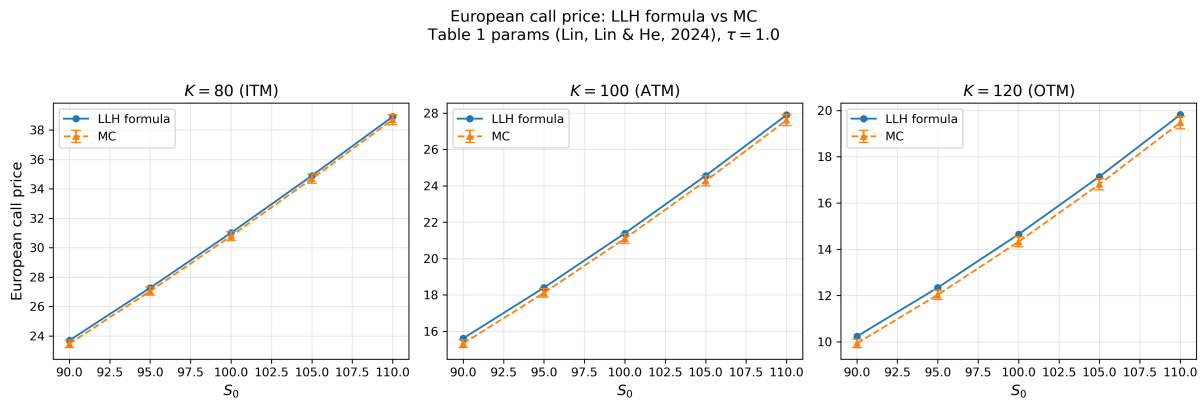

Figure 1: LLH formula vs MC — Table 1, $\tau = 1.0$, 10^5 paths.

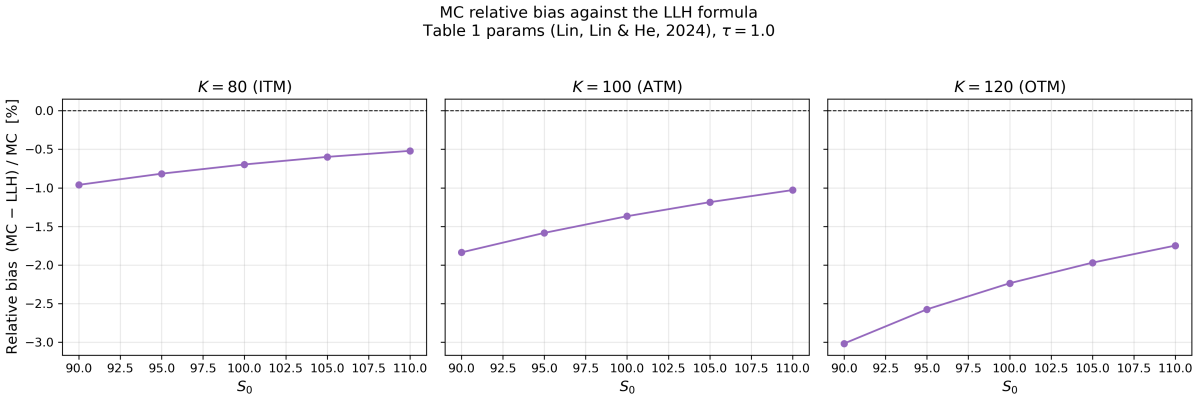

Figure 2: Relative error — Table 1.

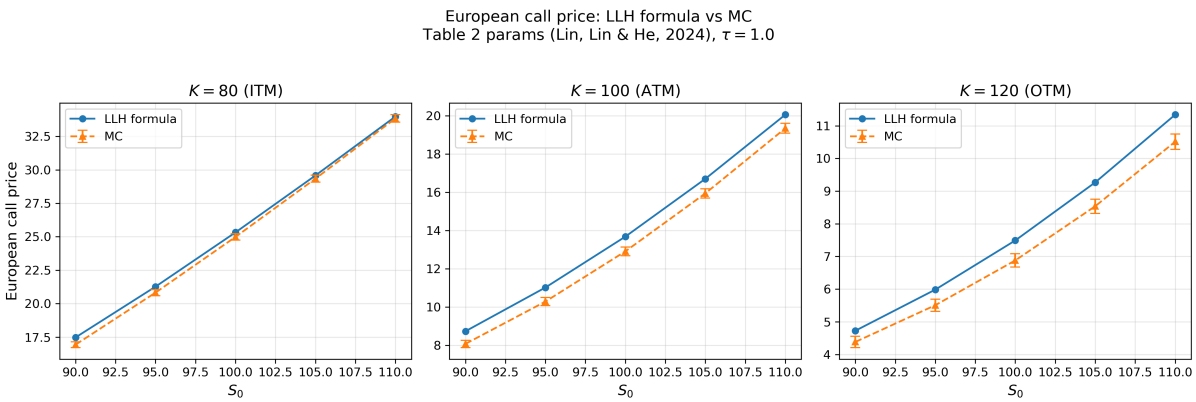

Figure 3: LLH formula vs MC — Table 2, $\tau = 1.0$, 10^5 paths.

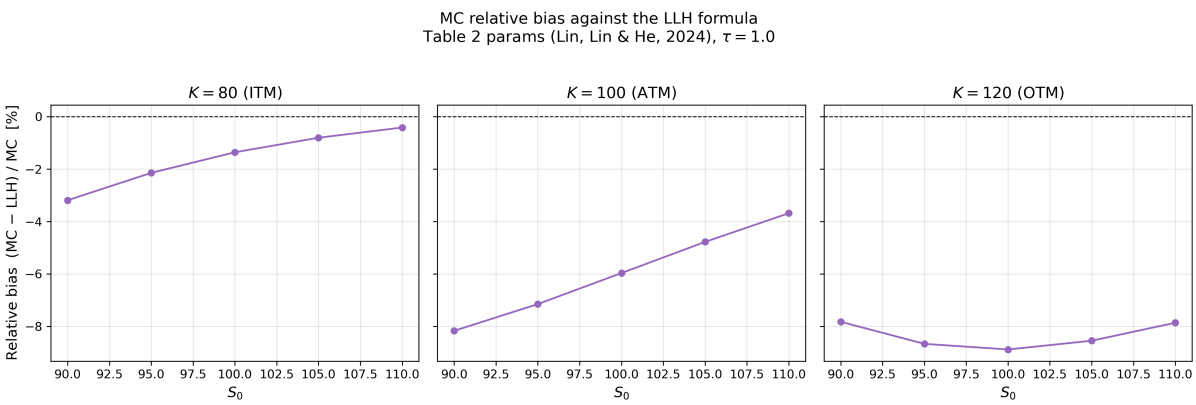

Figure 4: Relative error — Table 2.

Remarks.

- By put-call parity, $C_{\text{LLH}} - C_{\text{MC}} \approx P_{\text{LLH}} - P_{\text{MC}}$, so the absolute bias is comparable for calls and puts.
- Increasing number of MC simulations seems to reduce the error as show in the below figure. Increasing further the number of scenarios proved prohibitive.

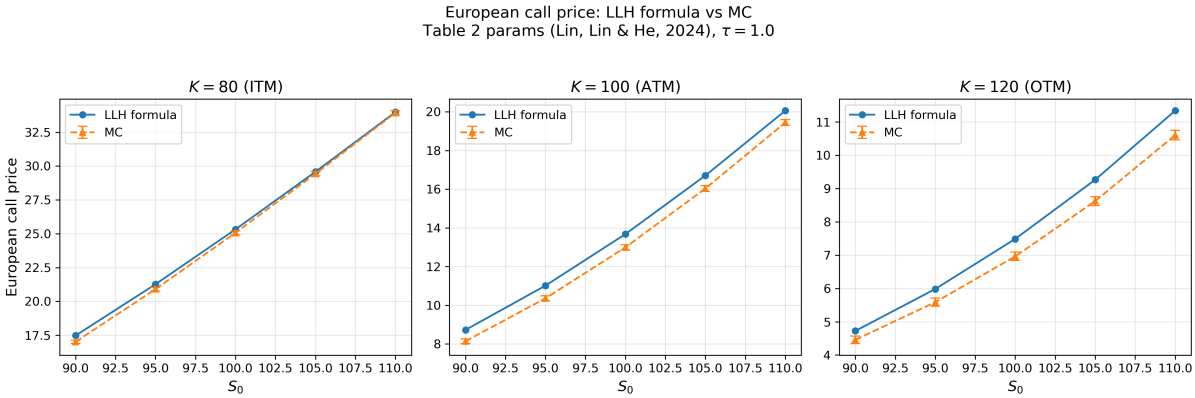

Figure 5: LLH formula vs MC — Table 2, $\tau = 1.0$, 5×10^5 paths.

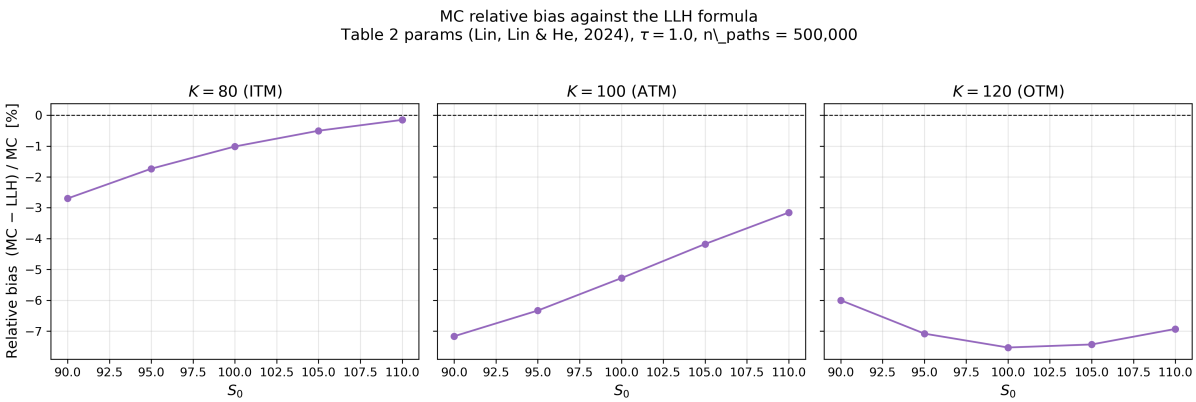

Figure 6: Relative error — Table 2.

5 Computational Performance

Table 2 compares the wall-clock cost of the LLH semi-analytical formula (ODE integration + trapezoid quadrature) against a Monte Carlo simulation with 500 000 paths. Timings are the median of five runs.

Horizon	LLH (s)	MC (s)	Speedup
1 month ($\tau = 1/12$, 22 steps)	$\sim$0.02	$\sim$0.5	$\sim$25$\times$
1 year ($\tau = 1.0$, 52 steps)	$\sim$0.02	$\sim$1.2	$\sim$60$\times$

Table 2: Computational cost: LLH formula vs Monte Carlo. $S_0 = 100$, $K = 90$, LLH Table 1 parameters.

The semi-analytical formula is significantly faster than Monte Carlo, with the speedup growing for longer maturities (more MC time steps). This makes it the preferred method for European pricing.

6 Edge-Case Validation

The LLH model nests three classical models as special cases. We verify that the implementation recovers the known prices in each limit, see notebook here.

6.1 Black–Scholes Limit ($\kappa = \nu = \lambda = \eta = \rho = 0$)

Setting all stochastic volatility parameters to zero reduces the LLH model to geometric Brownian motion with constant volatility σ_0. In this limit:

- The LLH formula price matches the Black–Scholes closed form to machine precision.
- Simulated terminal prices pass the Kolmogorov–Smirnov test for lognormality, confirming correct path generation.

Parameters: $r = 0.05$, $\sigma_0 = 0.2$, $\theta_0 = 0$, $S_0 = 100$, $K = 90$, $\tau = 1.0$.

6.2 Stein–Stein Limit ($\lambda = \eta = \rho = 0$)

Setting $\lambda = \eta = 0$ removes the time-varying θ_t dynamics, and $\rho = 0$ eliminates the leverage effect, recovering the original Stein & Stein (1991) model.
Parameters: $r = 0.0953$, $\kappa = 4.0$, $\nu = 0.1$, $\sigma_0 = 0.2$, $\theta_0 = 0.2$, $S_0 = 100$, $K = 100$, $\tau = 1/12$.

The LLH formula returns a call price consistent with the Stein–Stein Table 1 reference value of 10.77.

6.3 Schöbel–Zhu Limit ($\lambda = \eta = 0$)

Setting $\lambda = \eta = 0$ while allowing $\rho \neq 0$ recovers Schöbel & Zhu (1999). We reproduce their Table 2, computing European call prices across $\rho \in \{0.5, 0.0, -0.5\}$, $\theta_0 \in \{0.0, 0.1, 0.2, 0.3\}$, and $K \in \{90, 95, 100, 105, 110, 115, 120\}$ with $S_0 = 100$, $\tau = 0.5$, $r = 0.0953$, $\kappa = 4$, $\nu = 0.1$, $\sigma_0 = 0.15$.

The LLH formula prices match the published S&Z values to four decimal places in all tested configurations.

7 Further Validation of the Implementation

We test the implementation against properties expected of the pricing formula under the LLH dynamics. The formula must price the impact of λ and η on volatility in a consistent fashion. For example, all parameter values equal, a positive drift is expected to price higher than S&Z model. The implemented formula exhibits consistent pricing across the sensitivity tests to λ and η.

7.1 Effect of LLH Extensions over the Schöbel–Zhu Limit

Setting $\lambda = \eta = 0$ recovers the Schöbel–Zhu (1999) dynamics with constant long-run mean. We observe that for LLH Table 1 parameters the λ-drift ($\lambda = 0.9$) substantially increases option prices across all spots, reflecting a rising long-run mean that pushes the future volatility distribution upward. For the LLH Table 2 parameters the effect is smaller, consistent with the moderate $\lambda = 0.3115$.

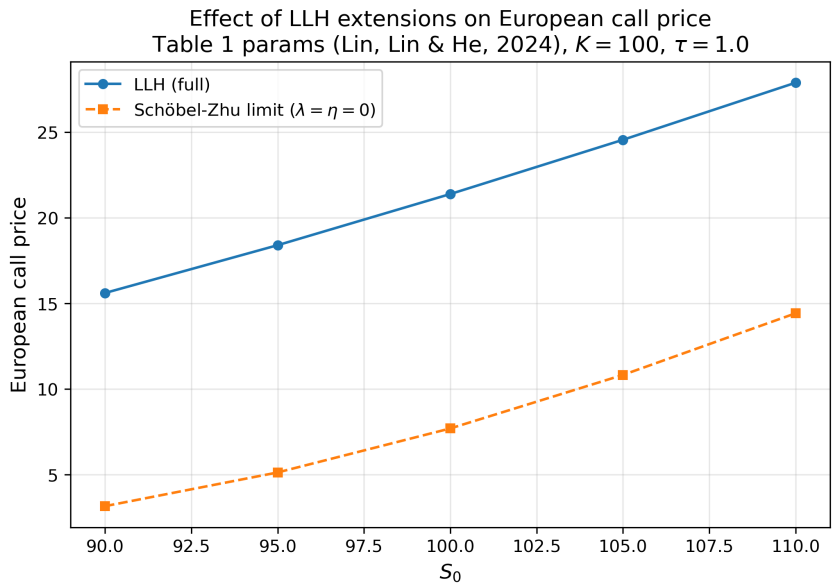

Figure 7: Full LLH vs S&Z limit — Table 1, $K = 100$, $\tau = 1.0$.

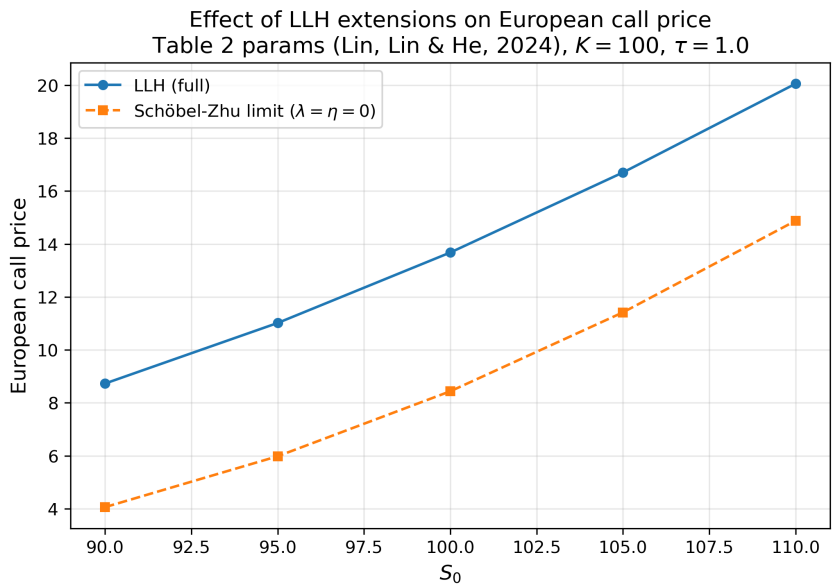

Figure 8: Full LLH vs S&Z limit — Table 2, $K = 100$, $\tau = 1.0$.

7.2 Sensitivity to λ and η

The parameters λ and η are the two additions of the LLH model relative to the Schöbel–Zhu framework.

7.2.1 Sensitivity to λ

We sweep $\lambda \in [-1, 1]$ while holding all other parameters fixed. As expected, the price is convex in λ with the minimum near $\lambda = 0$ and positive λ has a stronger effect than negative λ of the same magnitude. The sensitivity is most pronounced for OTM options ($K = 120$). Notice how larger η in LLH Table 2 parameters exacerbates the price increase over the S&Z benchmark.

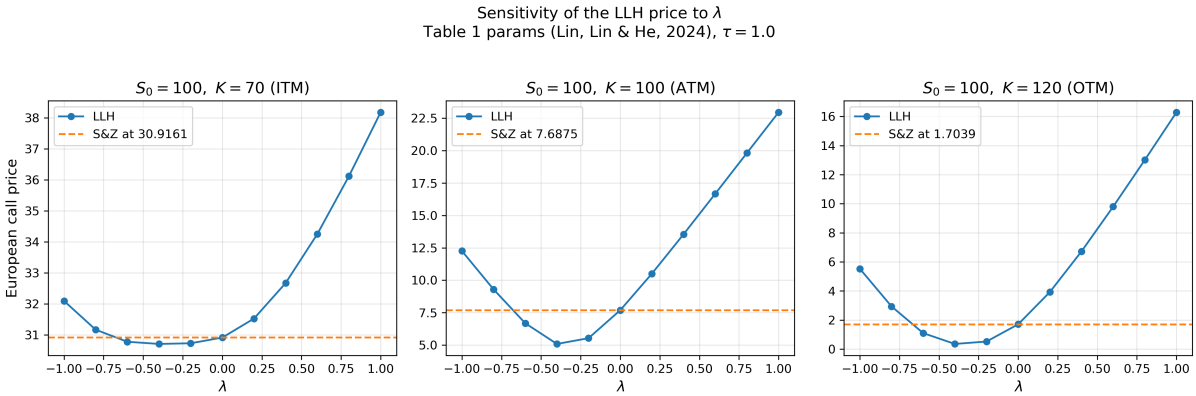

Figure 9: LLH price vs λ — Table 1, $\tau = 1.0$. Dashed: S&Z price.

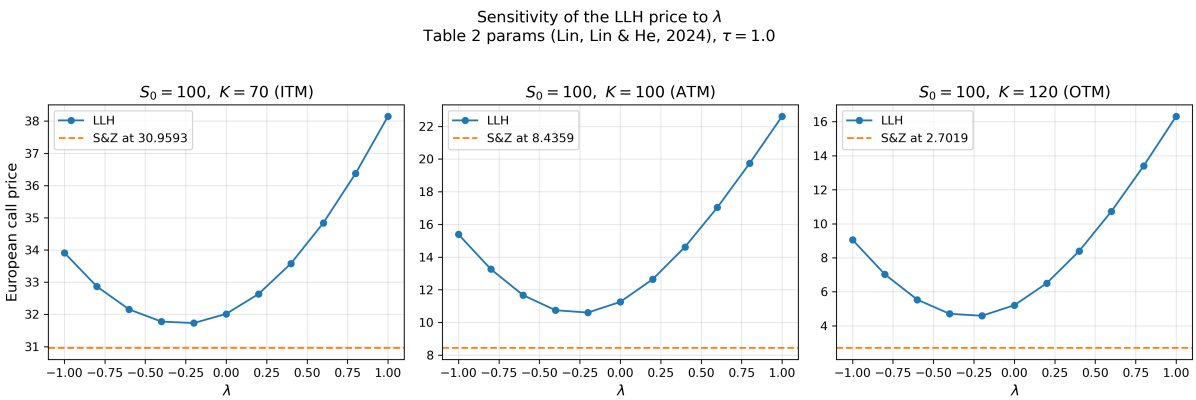

Figure 10: LLH price vs λ — Table 2, $\tau = 1.0$. Dashed: S&Z price.

7.2.2 Joint sensitivity to λ and η

We overlay curves for $\eta \in \{0.1, 0.15, 0.2\}$, each sweeping $\lambda \in [-1, 1]$. As desired, higher η amplifies sensitivity to λ: η controls the diffusion of θ_t, and larger diffusion widens the distribution of future volatility levels.

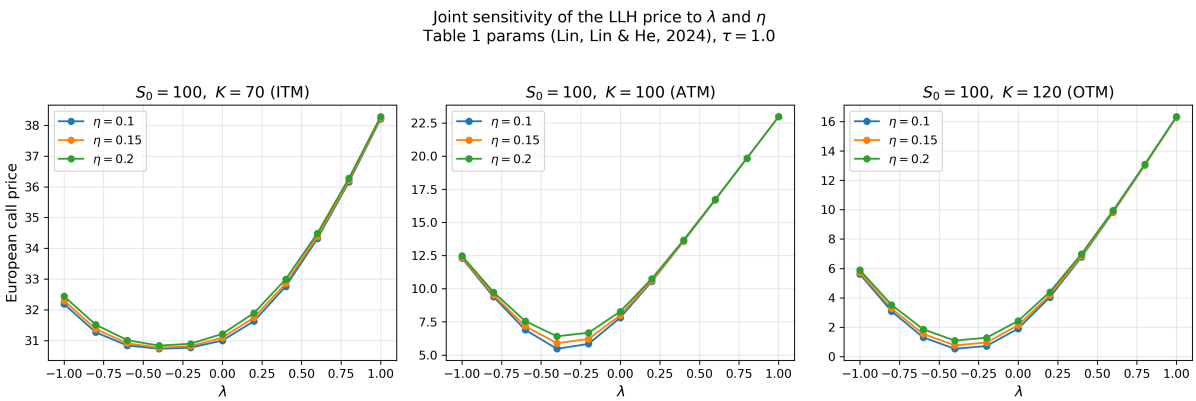

Figure 11: Joint λ–η sensitivity — Table 1, $\tau = 1.0$.

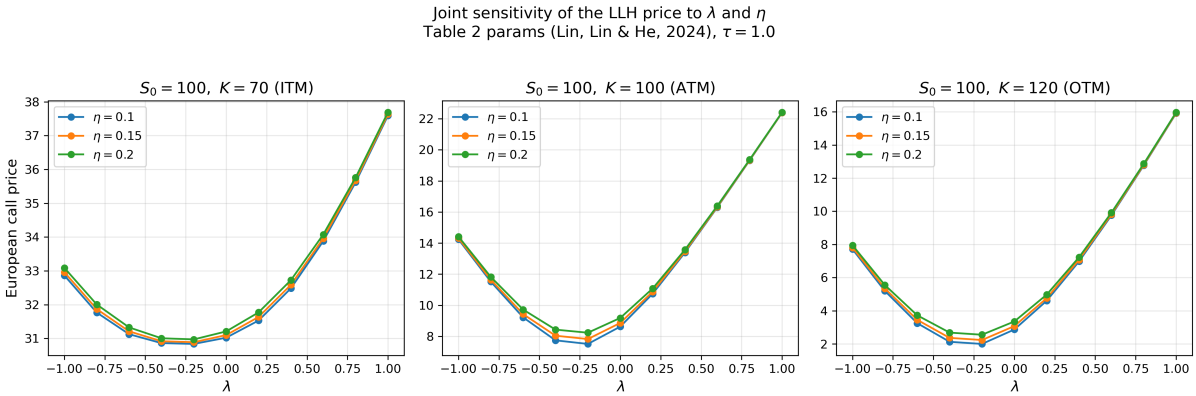

Figure 12: Joint λ–η sensitivity — Table 2, $\tau = 1.0$.

8 Conclusion

We have tested our implementation of LLH pricing formula against the prices of edge case models evaluated independently under different sets of parameters. The results show agreement to within four decimal digits consistently across models. The implemented LLH price has shown accurate to within 3% of the MC price for ITM options and up to within %7.5 for OTM options thus establishing the validation of the implementation. The higher discrepancy for OTM options results from their lower price making relative error more sensitive to simulation error (which we found prohibitive to illustrate). However, increasing the number of simulation paths from 10^5 to 5×10^5 suggests that it may be possible to reduce the error margin for this moneyness regime (error down %1.5).

A sensitivity analysis across the λ and η parameters reveals that the implemented LLH formula is consistent in the pricing across a sweep of volatility regimes. Furthermore, the LLH pricing formula demonstrates a considerable gain in speed at x60 times for 1 year maturity with 52 weekly steps. This suggest that the LLH formula is the recommended method for pricing under the LLH dynamics.

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